

MAHESWARA REDDY

Staff Engineer | SIP / VoIP | FreeSWITCH | C++ | Golang | Telecom & Media

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PROFESSIONAL SUMMARY

Staff Engineer with 5+ years of software engineering experience across **SIP/VoIP, telecom, FreeSWITCH, real-time media, C++, and Golang**. Experienced in developing SIP-based communication and IVR solutions, call-routing components, media-server functionality, and deployment automation. Strong hands-on background in SIP signaling, RTP, ESL, MRCP, GStreamer, Linux, Docker, Ansible, and CI/CD, with experience contributing to client-focused POCs and high-concurrency communication solutions.

TECHNICAL SKILLS

Programming: C, C++, Golang, Python, Shell Scripting

Telecom / VoIP: SIP, VoIP, FreeSWITCH, ESL, IVR, MRCP, SDP, RTP/RTCP, SIP over SCTP

Networking: TCP/IP, UDP, HTTP, WebSockets, WSS, SCTP, Socket Programming

Media / WebRTC: GStreamer, WebRTC, H.264, VP8, VP9, SFU/MCU, Audio/Video Encoding & Decoding

DevOps / Automation: Docker, Ansible, Rio CI/CD Pipelines, Infrastructure Automation

Tools: Git, JIRA, Radar, GDB, Valgrind, Wireshark, Tcpdump, SIPp

Build / Data / OS: Linux, Make, CMake, PostgreSQL, Multithreading, Data Structures

PROFESSIONAL EXPERIENCE

Insemi Technology Pvt. Ltd. (an Infosys company)

Staff Engineer | May 2026 – Present

Promoted from Staff Software Engineer

Staff Software Engineer | Dec 2024 – Apr 2026

- Contributed to the design and development of **SIP/VoIP-based communication and IVR solutions using FreeSWITCH** for scalable call-center and real-time communication use cases.
- Independently developed the **FreeSWITCH ESL and Golang-based call-routing and event-driven component**, supporting dynamic call-flow and routing requirements.
- Worked across **SIP signaling, FreeSWITCH dialplans, ESL, IVR call flows, RTP/media handling, and MRCP integrations** for client-focused POCs and solution development.
- Collaborated with clients and cross-functional engineering teams to understand requirements, troubleshoot call-flow and integration issues, and deliver functional POCs demonstrating solution feasibility.
- Independently designed and implemented **Ansible automation and deployment playbooks** for containerized applications and integrated deployment workflows with **Docker and Rio CI/CD pipelines**.
- Contributed to performance, scalability, troubleshooting, and deployment activities, including testing and optimization of **high-concurrency FreeSWITCH-based solutions**.

BRRaySoft Solutions Pvt. Ltd. — Deputed to Infosys Pvt. Ltd.

Senior Consultant | May 2024 – Dec 2024

- Worked on **VoIP, SIP, and FreeSWITCH application development** for communication solutions.
- Contributed to SIP-based FreeSWITCH solutions, experimentation, and POC development.
- Worked with cross-functional teams to design and deliver POCs and contributed to POC development best practices and standards.

Mirafr Software Technologies, Hyderabad

Software Engineer II | Dec 2023 – May 2024

Worked on software development in the telecom/media domain, contributing to C++-based communication and media solutions.

Vectone Mobile India Services Pvt. Ltd., Chennai

Software Engineer – C++ | Jan 2021 – Oct 2023

- Developed telecom/media software using **C++** on Linux, with hands-on work in networking, SIP, RTP, GStreamer, and real-time media processing.
- Worked on media-server capabilities including **audio/video encoding, decoding, transcoding, stream routing, video mixing, and WebRTC**.
- Implemented and debugged communication features involving **DTLS, SIP re-INVITE, RTP, SCTP, TCP/IP, and JSON/TCP messaging**.

Vector India Embedded Training Institute, Chennai

Embedded Trainee | Aug 2019 – Jan 2020

Completed professional embedded software training covering C, C++, Embedded C, I2C, SPI, UART, and CAN.

SELECTED PROJECTS

Next-Gen Call Center Platform using FreeSWITCH

Designed and developed a scalable SIP-based FreeSWITCH solution for a next-generation call-center platform; integrated third-party APIs and contributed to a working POC. **Technologies:** FreeSWITCH, SIP, Golang, C/C++, Python, Shell, Ansible, Linux.

Media Processing & Dynamic Tile Arrangement

Implemented dynamic multi-stream video tile arrangement and media-processing logic with focus on concurrency, synchronization, stability, and error handling. **Technologies:** GStreamer, UDP, C/C++, Linux.

DTLS Reconnection Between Client and Media Server

Implemented reconnection handling using DTLS, socket notifications, SIP re-INVITE, and RTP negotiation to restore media connectivity after connection failures. **Technologies:** C++, SIP, DTLS, RTP, GStreamer.

SIP over SCTP

Worked on reliable transport of larger SIP/SDP messages using SCTP, addressing message ordering and loss considerations compared with TCP. **Technologies:** SIP, SCTP, TCP/IP, C/C++.

EDUCATION

B.Tech — KSRRM College of Engineering, Kadapa | 2019 | First Class

CERTIFICATIONS & ACHIEVEMENTS

- Professional Embedded Training — Vector India Embedded Training Institute (Aug 2019 – Jan 2020).
- Won 3rd Prize in a Digital India presentation at NBKR College of Engineering, Nellore.

LANGUAGES

English — Read / Write / Speak Telugu — Read / Write / Speak